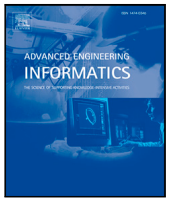
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3D real-time dynamic path planning for UAV based on improved interfered fluid dynamical system and artificial neural network

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ABSTRACT

In complex and volatile unknown flight environments, the limited environmental information obtained by sensors in the face of sudden dynamic and static obstacles makes it extremely challenging for unmanned aerial vehicles (UAVs) to obtain a safe and efficient path to avoid obstacles and reach a designated target point. Therefore, a real-time dynamic path planning method based on an improved interfered fluid dynamical system (IFDS) and artificial neural network (ANN) is proposed to enhance path quality and computational efficiency. Firstly, to address the issue of insufficient sample quality and quantity, IFDS is employed as the fundamental method for path planning to simulate and generate an adequate amount of sample data for the ANN training. Then, an enhanced sand cat swarm optimization algorithm (ESCSO) with an adaptive social neighborhood search mechanism and Lévy flight strategy is proposed to improve the sample quality. Secondly, the information between the UAV and the target points and obstacles is extracted from the sample data as the input for the network, the parameters of the IFDS are used as the feature extraction at the output of the network, and the ESCSO is applied to optimize the weights and biases of the ANN, enabling offline training of the neural network. Finally, the trained neural network is utilized to dynamically output IFDS parameters based on the real-time environmental information obtained from the sensors, enabling the generation of real-time obstacle avoidance paths. Experimental results in a series of complex simulated environments demonstrate that the proposed method outperforms other algorithms in terms of path quality and meets real-time requirements. It provides excellent obstacle avoidance characteristics for the UAV.

1. Introduction

In recent years, unmanned aerial vehicles (UAVs) have experienced significant development and widespread applications in both military and civilian domains, due to their excellent maneuverability, strong survivability, and ability to perform high-risk missions [1]. In military applications, UAVs are utilized for ground attacks, border surveillance, target identification, and other military tasks [2]. In the civilian sector, UAVs are employed for operations such as power inspections, resource exploration, and search and rescue operations [3]. The continuous advancements in artificial intelligence and computer technology over the past few decades have propelled the development of UAV technology, representing the forefront of modern high-tech advancements. As an autonomous unmanned control system, UAVs are characterized by intelligence and autonomy, which are the basic attributes of their development and future trends [4,5].

Path planning is a critical component for enhancing the autonomous flight capabilities of UAVs. Traditional path planning searches for globally optimal paths by offline search and is commonly used in static or known environments. However, the actual flight environment of UAVs is usually dynamic and unknown. Path planning in three-dimensional dynamic environments presents several challenges. For instance, various dynamic or sudden static obstacles in complex environments pose threats to the UAV's flight path, increasing uncertainty and complexity in path planning. Additionally, for flight environments with dense obstacles, sensor acquisition of precise obstacle locations will be extremely limited, and path planning algorithms need to be efficient and responsive in making decisions to adapt to dynamic changes in the environment to ensure flight safety. Finally, the balancing of multiple constraints is another difficulty in path planning. The path planner needs to consider constraints such as obstacle avoidance, minimum flight time, maximum turn angle, and maximum climb angle, which can sometimes conflict, requiring careful consideration and decision-making while ensuring safety. The above challenges make

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path planning for UAVs highly complex and demanding. Therefore, it is inevitable to develop an effective real-time path planning approach.

Specifically, the objective of real-time path planning refers to obtaining optimal control commands based on real-time environmental information detected by sensors equipped on the UAV (e.g., forward-looking sonar, high-frequency radar, etc.) to autonomously generate safe paths for the UAV that satisfy various constraints. It is important and necessary for these constraints in practical applications. For instance, the obstacle avoidance constraint ensures that the UAV avoids collisions with other flying objects or obstacles during flight. The minimum flight time constraint aims to minimize mission duration while ensuring safety. Constraints such as maximum turn angle and maximum climb angle are vital for flight performance and safety considerations, ensuring the UAV maintains stability and safety during flight. Therefore, a robust real-time path planner can effectively address the challenges of safe UAV flight in complex environments and take into account critical constraints, which enhances the autonomy and usability of UAVs in dynamic and variable environments. Over the past decades, numerous scholars have proposed methods to tackle path planning problems in dynamic environments. For example, Pu et al. proposed a double adaptive A* algorithm to improve real-time path planning speed and validated its advantages over traditional algorithms through simulation experiments [6]. Huang et al. presented a multi-robot navigation path planning method called PONA based on the generalized Voronoi diagram, which achieves collision-free and efficient navigation in space through priority ordering and path optimization, resulting in shorter path lengths compared to traditional algorithms [7]. However, the aforementioned methods are all based on map search algorithms that require pre-generation and storage. For larger search areas, this can lead to considerably slower planning speeds. In [8], a real-time path planner based on probabilistic constraints and the rapidly exploring random tree (RRT) was introduced to address probabilistic robust planning problems in uncertain dynamic environments. Despite its ability to yield favorable results, this method proves inefficient for planning within narrow flight regions, occasionally leading to non-feasible solutions. In [9–12], many scholars have adopted the artificial potential field method (APF) or its variants to solve the problem of dynamic path planning for UAVs. Although these methods have exhibited promising outcomes within specific simulation environments, they are heavily reliant on parameter settings. This reliance renders the process of tuning relatively intricate, demanding experience and iterative experimentation. These limitations make it difficult to effectively guarantee the effectiveness and reliability of the algorithms in complex environments and practical applications.

Swarm intelligence algorithms have been widely applied in path planning problems due to their simple structure, ease of implementation, and excellent optimization performance. Utkarsh Goel et al. proposed a glow-worm swarm optimization-based path planning algorithm to find collision-free paths for UAVs in a three-dimensional dynamic environment, and its superiority is validated through simulation experiments [13]. Zhang et al. presented a predator–prey pigeon-inspired optimization algorithm (PPPIO) to deal with the three-dimensional path planning problem of UAVs in dynamic environments. By comparing simulation results, the efficiency and superiority of the PPPIO algorithm in solving the UCAV three-dimensional path planning problem are validated [14]. In [15], the paper introduced an online path planning approach based on teaching-learning-based optimization (TLBO), suitable for autonomous tracked vehicles in cluttered environments. The method takes into account factors like path smoothness, collision avoidance, and vehicle dynamic constraints, with experimental results showcasing its effectiveness. Seyedhadi et al. proposed a novel approach based on the cuckoo optimization algorithm to tackle path planning issues for mobile robots within a dynamic environment [16]. The method finds a short-length, secure, smooth, and collision-free path while reducing computational complexity through an optimized feature vector. Despite the effectiveness of swarm intelligence algorithms highlighted in the aforementioned studies, their iterative optimization approach for obtaining obstacle avoidance paths restricts them from generating real-time safe paths when encountering sudden dynamic or static obstacles. As a result, they fail to meet the requirements of real-time obstacle avoidance. For multi-agent systems, task allocation and path planning are two significant research directions. In [17], Chen et al. proposed a novel scheduling algorithm aimed at shortening the workflow through modeling and optimization methods. Experimental results confirmed the reliability and efficiency of this approach. In [18,19], Chen et al. investigated the problem of path planning for multi-heterogeneous UAVs in coverage tasks. They proposed an adaptive clustering-based path planning algorithm and an ant colony system algorithm to select suitable paths for each UAV and effectively cover all regions. While experimental results verified the performance enhancements and reduced time consumption and energy wastage in both of the above methods, these approaches did not consider path planning issues for UAVs in dynamic scenarios.

To enhance the autonomy of obstacle avoidance systems, some researchers adopt reinforcement learning and neural networks to boost obstacle avoidance capabilities. For instance, Ye et al. proposed a neural fuzzy system with mixed learning phases to achieve efficient obstacle avoidance [20]. This system combines supervised learning and reinforcement learning to automatically learn fuzzy rules, facilitating collision-free navigation for mobile robots. Zhang et al. presented a fusion scheme in which a collision-free reference path for UAVs is generated through static path planning, followed by dynamic collision avoidance utilizing a Markov decision process [21]. The effectiveness of the method is validated through simulation experiments. Chen proposed a dynamic obstacle avoidance path planning method based on the soft actor-critic (SAC) algorithm [22]. By designing a comprehensive reward function and employing a prioritized experience replay technique, it achieves real-time avoidance of moving obstacles and a high success rate in planning tasks. Chen et al. proposed a reinforcement learning algorithm based on global and local attention to address the challenges of UAVs in cooperative behavioral control for solving the decision-making problem in multi-UAV cooperative tasks and verified the effectiveness and efficiency of the method through simulation experiments [23]. Despite significant progress in research on obstacle avoidance planning methods based on reinforcement learning, there are still challenges in designing reward functions and action policies for the algorithms. Even the most reasonable rewards cannot completely avoid the problem of local optima. Especially when UAVs fly in narrow environments, it is extremely difficult to design an optimal collision avoidance mechanism through reinforcement learning based on limited local environment information.

Furthermore, Lin et al. presented an online obstacle avoidance planning method for unmanned underwater vehicles (UUVs) based on CW-RNN and LSTM [24]. The experimental results demonstrate that the proposed method exhibits excellent performance in path planning and obstacle avoidance across diverse environments. Zhu presented a data-driven approach for multi-robot motion planning in dynamic environments [25]. It utilizes recurrent neural networks (RNNs) for trajectory prediction and incorporates them into a decentralized model predictive control (MPC) framework to achieve collision avoidance. Liu et al. presented a real-time path planning method for UAVs based on residual convolutional neural networks [26]. It aims to generate feasible paths for UAVs with limited environmental information. The experimental results demonstrate that the proposed method can provide safe obstacle avoidance paths. Although the aforementioned methods have achieved some results, the neural network-based path planning methods encounter challenges in obtaining high-quality and sufficient training samples, which directly affects the efficiency of path planning. Additionally, it is difficult to obtain a network model with better weights, which compromises the performance and accuracy of path planning tasks.

In recent years, inspired by the natural phenomenon of water in rivers smoothly avoiding rocks and reaching its destination, Wang et al. have proposed the interferential fluid dynamics system (IFDS) computational method that can meet the requirements of real-time path planning and obstacle avoidance in three-dimensional complex environments [27]. Compared to other algorithms, it has the following advantages. Firstly, IFDS is a dynamic path planning method with real-time decision-making capability. It can dynamically adjust paths based on the current positions of obstacles and their dynamic changes in the environment. This means that in three-dimensional space, the UAV can make timely decisions to avoid collisions or disturbances. Secondly, IFDS exhibits a high degree of adaptability and can accommodate various types of obstacles and changing environments. This adaptability allows it to perform exceptionally well in various scenarios in three-dimensional space. Meanwhile, IFDS employs a fluid dynamics model to simulate the obstacle avoidance process in the flight environment. This model generates smooth paths without the need for additional overhead, enabling the UAV to navigate efficiently in three-dimensional space. Moreover, efficiency is another notable strength of IFDS. By quantifying the influence of obstacles on the flow field through disturbance matrices, IFDS can efficiently compute paths, ensuring the UAV avoids obstacles with minimal energy consumption in three-dimensional space. However, IFDS has some parameters that can impact the quality of the paths. Therefore, in complex dynamic environments with various obstacles, it is particularly important to adaptively adjust the parameters to generate high-quality paths.

Based on the above analysis, this paper proposes a real-time path planning method for UAVs based on improved IFDS and artificial neural networks (ANNs) to address the shortcomings. Firstly, this paper adopts the IFDS algorithm as the path planning module and leverages the powerful autonomous learning capability and high real-time responsiveness of neural networks to address the problem of adaptive variation of IFDS parameters in dynamic environments. Secondly, to address the issue of insufficient training samples for the ANN, this paper constructs complex dynamic simulation scenarios and utilizes the IFDS algorithm to generate training samples. Then, the enhanced sand cat swarm optimization algorithm with adaptive social neighborhood search mechanism and Lévy flight strategy (ESCSO) is proposed to optimize the quality of the samples. In addition, environmental state information and IFDS parameter values are extracted from the sample data, and the ESCSO algorithm is applied to optimize the parameters of the ANN network, which is subsequently trained offline. Finally, the environmental state information of the new scenario obtained from the onboard sensors is fed into the trained network model, and the predicted parameter values are input into the IFDS algorithm, generating real-time UAV behavior as guidance for path planning.

The primary contributions of this study are demonstrated as follows: (1) Based on IFDS, a novel fluid-based three-dimensional obstacle avoidance framework is proposed by integrating autonomous learning and real-time responsive ANN. This framework can realize real-time autonomous path planning under the condition that agents rely solely on the local information detected by its onboard sensors, significantly enhancing the quality and real-time capabilities of path planning. (2) Within the framework described above, a novel approach is introduced to address the issues of low-quality and insufficient training samples for ANN. Firstly, complex dynamic simulation scenarios are constructed, with IFDS serving as the foundational path planning method. Secondly, the modified sand cat swarm optimization algorithm (ESCSO) with an adaptive social neighborhood search mechanism and Lévy flight strategy was proposed to optimize IFDS parameters for generating high-quality samples. This method provides a conducive offline training environment for neural networks. (3) To prevent neural networks from easily converging to local minima, the parameters of the ANN are optimized for the first time using the ESCSO algorithm to enhance the network's predictive capability for environmental state information. This process enables the neural network to more accurately predict the values of IFDS parameters, thereby improving the ability of the path planner to avoid obstacles in complex and dense environments. The experimental results indicate that the proposed algorithm exhibits higher robustness and stronger real-time performance compared to other algorithms, while the planned paths achieve superior performance metrics.

The remainder of this thesis is structured as follows: Section 2 provides the Preliminary. The proposed real-time path planning algorithm is introduced in Section 3. The simulation findings are presented in Section 4. Section 5 gives the discussion analysis. Finally, the conclusion is furnished in Section 6.

2. Preliminary

2.1. Obstacle environment modeling

In complex flight environments, UAVs often encounter various types of obstacles, and high-quality modeling of different types of obstacles is a prerequisite and foundation for path planning tasks. To achieve a good balance between accuracy and efficiency, a unified standard model is adopted to represent various irregular obstacles. The formulation for the model is expressed as follows [28]:

$$F(x,y,z)=\left(\frac{x-x_c}{e}\right)^{2m}+\left(\frac{y-y_c}{f}\right)^{2n}+\left(\frac{z-z_c}{g}\right)^{2q}=1, \tag{1}$$

where (x_c, y_c, z_c) is the center position coordinates of the obstacle; m, n, q and e, f, g are positive numbers that define the size and geometry of the obstacle. For example, when $m = n = q = 1$ and $e = f = g$, the obstacle is a sphere; when $m = n = 1, q > 1$ and $e = f$, the obstacle is a cylinder. $F > 1, F = 1$, and $F < 1$ respectively indicate that the UAV position is located outside, on the surface, and inside the equivalent envelope of the obstacle.

2.2. IFDS algorithm

The IFDS algorithm mimics the characteristics of water flow in nature and constructs the fundamental model of path planning. When there are no obstacles, the flow of water moving in a straight line is considered as the initial flow field. When encountering obstacles, the water flows smoothly bypassing the obstacles is considered as the disturbed flow field [28]. Assuming that the current position of the UAV is $P(x, y, z)$ and the destination position is $E(x_e, y_e, z_e)$, with the desired flight speed of V_{UAV}. When there are no obstacles in the environment, the initial flow field (flight path) should be a straight line from P to E, and the initial flow velocity (flight speed vector) in the inertial frame should be:

$$\sigma=\left(\frac{V_{UAV}(x-x_e)}{dis(P,E)},\frac{V_{UAV}(y-y_e)}{dis(P,E)},\frac{V_{UAV}(z-z_e)}{dis(P,E)}\right)^T, \tag{2}$$

$$dis(P,E)=\sqrt{(x-x_e)^2+(y-y_e)^2+(z-z_e)^2}. \tag{3}$$

If there are m obstacles in the flight environment, the influence of obstacles on the initial flow field is represented by a perturbation matrix R. It is defined as follows:

$$R=\sum_{i=1}^{m}\beta_i R_i, \tag{4}$$

where β_i denotes the coefficient of the kth obstacle, whose value is calculated as follows:

$$\beta_i'=\begin{cases}1, & if\ m=1\\ \prod_{i=1,i\neq m}^{m}\frac{F_i-1}{(F_i-1)+(F_i-1)}, & otherwise\end{cases}, \tag{5}$$

$$\beta_i=\frac{\beta_i'}{\sum_{i=1}^{m}\beta_i'}. \tag{6}$$

R_i represents the perturbation matrix of the ith obstacle, which is defined as follows:

$$R_i=I-\frac{Q_iQ_i^T}{|F_i|^{\frac{1}{\delta_i}}|Q_i|^2}+\frac{S_iQ_i^T}{|F_i|^{\frac{1}{\varphi_i}}|S_i||Q_i|}, \tag{7}$$

where I indicates the third order unit matrix; $-\frac{Q_i Q_i^T}{|F_i|^{\frac{1}{\delta_i}}|Q_i|^2}$ and $\frac{S_i Q_i^T}{|F_i|^{\frac{1}{\varphi_i}}|S_i||Q_i|}$ refer to the repulsion and tangential matrices, respectively. $Q_i = \left[\frac{\partial F_i}{\partial x}, \frac{\partial F_i}{\partial y}, \frac{\partial F_i}{\partial z}\right]^T$ represents the radial normal vector perpendicular to the kth obstacle outward, which is the vector of partial derivatives of the obstacle function F. $\delta_i = \delta \cdot e^{(1-\frac{1}{dis \cdot L})}$, where δ represents the repulsive coefficient; L is the distance between the UAV and the surface of the obstacle. $\varphi_i = \varphi \cdot e^{(1-\frac{1}{dis \cdot L})}$, where φ is the tangential coefficient. S_i is a tangent vector in an arbitrary direction within the tangent plane of the ith obstacle, and it can be computed as follows.

S_i^1 and S_i^2 are defined as two mutually orthogonal tangent vectors. The tangent coordinate system $O_{x'y'z'}$ is constructed by using S_i^1, S_i^2, and Q_i as the x', y', and z' axes, respectively.

$$\begin{cases} S_i^1 = \left[\frac{\partial F_i}{\partial y}, -\frac{\partial F_i}{\partial x}, 0\right]^T \\ S_i^2 = \left[\frac{\partial F_i}{\partial x}\frac{\partial F_i}{\partial z}, \frac{\partial F_i}{\partial y}\frac{\partial F_i}{\partial z}, -(\frac{\partial F_i}{\partial x})^2 - (\frac{\partial F_i}{\partial y})^2\right]^T \end{cases} \quad (8)$$

Hence, any tangent vector in the tangent plane can be represented as follows:

$$S_i' = \left[cos\alpha_i, sin\alpha_i, 0\right]^T \quad , \quad (9)$$

where $\alpha_i \in [-\pi, \pi]$ represents the tangent direction coefficient, which is the rotation angle along the z' axis. In the inertial coordinate system O_{xyz}, the transformation of S_i' to S_i can be achieved using the following formula:

$$S_i = Rot_i S_i' \quad , \quad (10)$$

where Rot_i is the transformation matrix from $O_{x'y'z'}$ coordinate system to O_{xyz} coordinate system.

Then, considering the influence of the moving threat, the perturbed flow velocity $\bar{\sigma}$ can be obtained by modifying σ using R:

$$\bar{\sigma} = R(\sigma - Obs_v) + Obs_v \quad , \quad (11)$$

where Obs_v is the velocity vector of the obstacle, which is defined as follows:

$$Obs_v = \sum_{i=1}^{m} \beta_i e^{\frac{-(F_i-1)}{\phi_i}} Obs_{i,v} \quad , \quad (12)$$

where ϕ_i is a constant.

Finally, the next flight path point of the UAV can be calculated by:

$$P_{UAV}^{t+1} = P_{UAV}^{t} + \bar{\sigma} \cdot \triangle T \quad , \quad (13)$$

where $\triangle T$ denotes the sampling period.

2.3. Multi-layer perceptron

The feedforward neural network represents the most common and widely applicable structure of ANNs, comprising numerous neurons distributed across each layer [29–31]. The input layer is present in the first layer of the feedforward neural network, the output layer in the last layer, and the hidden layer in the middle layer. When the feedforward neural network incorporates multiple hidden layers, it is referred to as a multilayer perceptron (MLP), as depicted in Fig. 1. In MLP, information propagates in a unidirectional manner, traversing through the input layer, the hidden layers, and culminating in the output layer. The implementation steps of MLP are illustrated in Eqs. (14)–(16).

Firstly, the input values of the hidden layer are acquired by applying the Eq. (14):

$$g_k = \sum_{j=1}^{N} \left(\mu_{j,k} \cdot D_j\right) + \varepsilon_k, k = 1, 2, \ldots, h \quad , \quad (14)$$

where N denotes the number of neurons in the input layer; $\mu_{j,k}$ is the weight coefficient; ε_k and D_k refer to the bias value of the kth neuron and the kth input data, respectively.

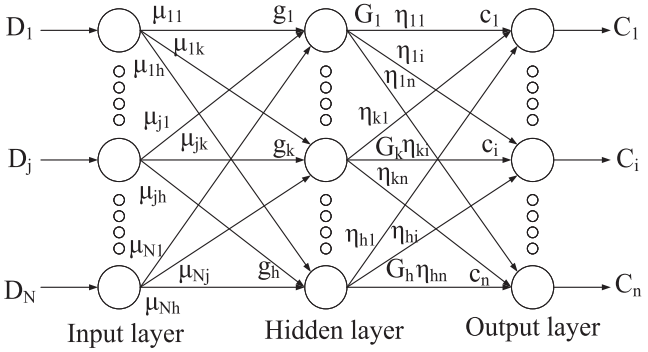


Fig. 1. Multilayer perceptron.

Secondly, the output values of the hidden layer are captured by employing the Eq. (15):

$$G_k = relu(g_k) = max(0, g_k), k = 1, 2, \ldots, h \quad , \quad (15)$$

Finally, the results of the output layer are given as follows:

$$\begin{aligned} c_i &= \sum_{k=1}^{h} \left(\eta_{k,i} \cdot G_k\right) + \varepsilon_i', i = 1, 2, \ldots, n \\ C_i &= tanh(c_i) = \frac{2}{1+e^{-2c_i}} - 1 \end{aligned} \quad , \quad (16)$$

where $\eta_{k,i}$ is the weight of the link between kth neuron and ith neuron; ε_i' and C_i denote the deviation value and output value of the ith neuron of the output layer, respectively.

2.4. Sand cat swarm optimization algorithm (SCSO)

The sand cat swarm optimization (SCSO) optimizer is a highly novel meta-heuristic algorithm designed by Amir Seyyedabbasi et al. in 2022 [32]. The algorithm mainly leverages the sensitive hearing of the sand cat to simulate its search and prey-catching behavior in order to obtain the optimal solution. The main actions of the SCSO algorithm and the corresponding model are introduced as follows.

2.4.1. Search for prey

Sand cats primarily prey on small desert animals such as gerbils, sandgrouse, and rabbits. In SCSO, individuals within the population are randomly selected as prey. The sand cat searches for prey based on its distance to the current target and its hearing range. The position update model for the sand cat is as follows:

$$\begin{aligned} \overrightarrow{P(t+1)} &= \vec{\xi} \cdot (\overrightarrow{P_{rand}(t)} - rand \cdot \overrightarrow{P(t)}) \\ \vec{\xi} &= \overrightarrow{\xi_G} \cdot rand(0, 1) \\ \overrightarrow{\xi_G} &= SR - (\frac{SL \times t}{max_iter}) \end{aligned} \quad , \quad (17)$$

where $\overrightarrow{P}$ represents the position vector of search agents in the swarm; t and max_iter indicate the current count of iterations and the maximum count of iterations of the algorithm, respectively; $\overrightarrow{P_{rand}}$ is the position vector of randomly selected search agent in the current swarm; SL is a constant and is set to 2.

2.4.2. Attacking prey

In the attack phase, the position update of the sand cat is mainly done with the participation of the current sand cat position and the optimal individual position in the population. Its specific attack model is portrayed as follows:

$$\begin{aligned} \overrightarrow{DD} &= \left|rand \cdot \overrightarrow{P_{best}(t)} - \overrightarrow{P(t)}\right| \\ \overrightarrow{P(t+1)} &= \overrightarrow{P_{best}(t)} - \vec{\xi} \cdot \overrightarrow{DD} \cdot cos(\vartheta) \end{aligned} \quad , \quad (18)$$

where ϑ is a random angle value among 0 and 360; $\overrightarrow{P_{best}}$ indicate the optimal search agent in the current swarm.

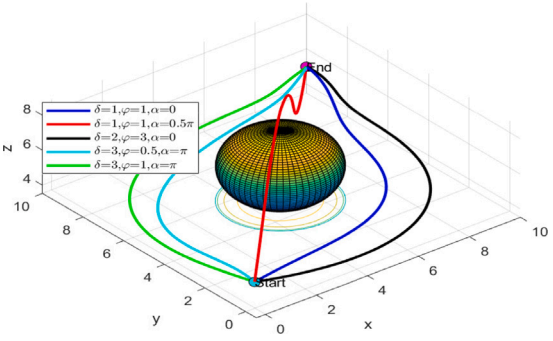


Fig. 2. Planning paths obtained by IFDS with different coefficients.

2.4.3. Bridging phase

The transition between exploration and exploitation in the algorithm primarily relies on the value of parameter ℓ. When $|\ell| > 1$, the algorithm performs the search phase; otherwise, it executes the attack phase. The specific model is presented as follows:

$$\ell = 2 \times \overrightarrow{\xi_G} \times rand - \overrightarrow{\xi_G}$$
$$\overrightarrow{P(t+1)} = \begin{cases} \vec{\xi} \cdot (\overrightarrow{P_{rand}(t)} - rand \cdot \overrightarrow{P(t)}), & if \ \ |\ell| > 1 \\ \overrightarrow{P_{best}(t)} - \vec{\xi} \cdot \overrightarrow{DD} \cdot cos(\vartheta), & otherwise \end{cases} . \tag{19}$$

3. Proposed real-time path planning algorithm

Compared to other path planners, the IFDS algorithm makes it a powerful tool for adapting UAV path planning in complex unknown dynamic environments due to its computational efficiency, handling of different shapes of obstacles, smooth planning paths, and real-time performance. However, IFDS also has some limitations. The specifics are described as follows:

From Eq. (7), it can be observed that the perturbation matrix R_i is dependent not only on the UAV's position and the obstacle function F, but also on two adjustable reaction coefficients δ and φ, as well as a directional coefficient α. The impact of these coefficients on the planned path is illustrated in Fig. 2.

As shown in Fig. 2, different combinations of coefficients can determine the shape and direction of the path. In complex environments, where UAVs encounter obstacles of diverse sizes and shapes, the parameters of the IFDS algorithm should adaptively adjust to yield robust, real-time, collision-free paths. In previous studies [33], the receding horizon control strategy was proposed to optimize the IFDS coefficients online. However, this method is time-consuming and cannot meet the real-time requirements of flight. Therefore, this paper proposes to combine MLP in ANNs with the IFDS algorithm and consider the IFDS algorithm as the base method for path planning, and the neural network trained by MLP is applied to optimize the corresponding path planning parameters for real-time path planning architecture. However, neural network-based approaches also have some shortcomings. In this paper, the ESCSO algorithm is proposed for generating high-quality data samples. Secondly, the ESCSO is used to apply to MLP and then obtain better weights and bias values. The specific operations are described in detail as follows.

3.1. Simulation samples generation based on ESCSO

3.1.1. Proposed enhanced sand cat swarm optimization algorithm (ESCSO)

Although SCSO demonstrates good performance compared to other algorithms such as PSO [34], GWO [35], and SSA [36] in some optimization problems, achieving the desired optimal solution with SCSO remains highly challenging. Therefore, two different operations are integrated into the SCSO algorithm to enhance its optimization capability. The specific operations are described in detail below.

(A) Adaptive social neighborhood search mechanism (ASNS)

The update mechanism of the SCSO algorithm indicates that the search prey phase controls the exploration ability of the algorithm and plays a crucial role in improving its search performance. Through the investigation of Eq. (17), it is found that the individual updating process is primarily accomplished with the participation of random individuals from the population, which effectively enhances the population diversity of the algorithm. However, the random selection mechanism carries certain blindness and lacks effective information exchange among the population, which affects the convergence speed of the algorithm. Moreover, the optimization accuracy of the algorithm is relatively modest when dealing with complex multimodal optimization tasks.

Hence, an adaptive social neighborhood search mechanism (ASNS) is presented to overcome these shortcomings. This strategy forms individual neighborhoods by calculating the social status and social influence of each individual in the population and establishing a social network. The SCSO utilizes this strategy to construct adaptive neighborhoods for sand cats and designs a novel update mechanism based on neighborhood search to search for prey. This not only enhances the mutual influence among individuals in the population but also effectively bolsters the diversity and computational accuracy of the algorithm. ASNS is composed of four primary operators: social ranking, assessing social impact, constructing social networks, and building neighborhoods. They are specifically defined as follows.

Social ranking: Firstly, based on the fitness of each individual in the current population, the sorted population is obtained by arranging them in ascending order of fitness. Then, according to Eq. (20), the social ranking of each individual is determined to establish their status within the current population:

$$SR_j = ZP + 1 - j, \quad j = 1, 2, \ldots, ZP, \tag{20}$$

where ZP and j denote the swarm scale and the index of individuals in the sorted swarm, respectively. From Eq. (20), it is clear that the better individuals in the swarm will receive a higher ranking value.

Assessing social impact: In the real world, individuals with varying social ranks exert attraction and influence over others within their social networks. Generally, individuals with higher-ranking values have a greater social influence on other individuals. Therefore, the evaluation of an individual influence is defined as follows:

$$SI_j = \frac{LL \cdot SR_j}{ZP}, \quad j = 1, 2, \ldots, ZP, \tag{21}$$

where LL is a constant and represents the social impact factor.

Constructing social network: After evaluating the social influence of each individual, a social network is constructed for each individual in the population using their social influence. Specifically, the relationship between P_j and P_a is established according to Eq. (22) as follows:

$$Rel(P_j, P_a) = \begin{cases} 1, & if \quad rand \le max(SI_j, SI_a) \\ 0, & otherwise \end{cases} . \tag{22}$$

From Eq. (22), it can be observed that individuals with greater social influence are more likely to have connections with a larger number of other individuals. Consequently, a higher number of individuals will have the chance to obtain better candidate solutions.

Building the neighborhood: Based on Eq. (22), if there is a connection between an individual and other individuals, they are considered neighbors. Therefore, the adaptive neighborhood of each individual can be constructed by examining their relationships with other individuals, and the specific operation is defined as follows:

$$Neighbors\ of\ P_j = \left\{ P_a / a \epsilon [1, ZP] \,\&\, j \neq a \,\&\, Rel(P_j, P_a) = 1 \right\}. \tag{23}$$

By combining ASNS with SCSO, a new search updating mechanism is proposed. When individuals in the population are in the search phase, the neighborhood information of the current individual is first extracted using ASNS. Then, based on the obtained neighborhood information, an

individual is selected as the prey. The specific updating mechanism is described as follows:

$$\overrightarrow{P_j(t+1)} = \vec{\xi} \cdot (\overrightarrow{P_k(t)} - rand \cdot \overrightarrow{P_j(t)}) \quad , \tag{24}$$

where $\overrightarrow{P_k(t)} \in Neighbors\ of\ \overrightarrow{P_j(t)}$.

(B) Lévy flight strategy

The attack prey phase of the SCSO controls the exploitation performance of the algorithm. Through the investigation of Eq. (18), it is found that the updating process of individuals is mainly influenced by the best individual in the current population, leading to the individuals' latest positions approaching the optimal solution. This greatly enhances the quality of candidate solutions in the algorithm. However, in dealing with complex multimodal optimization tasks, the SCSO algorithm suffers from local stagnation, resulting in low optimization accuracy. The reason behind this is that if the current optimal solution gets trapped in a local optimum, according to the current attack mechanism, the entire population will be stuck in a local extreme value. Relying solely on the ξ is insufficient to escape from local optima.

Lévy flight is a non-Gaussian random approach based on Lévy stable distribution for random walking [37,38]. Levy flight is characterized by random wandering with small step lengths and occasional directional abrupt jumps with larger step lengths, allowing it to search the same place without repetition [39,40]. The small step size increases the local search ability of the algorithm, and the occasional long step size enhances the ability of the algorithm to jump out of the local optimum. Metaheuristic algorithms utilize the random wandering and chaotic step lengths of Levy flight to enhance the walking capability of search agents in the search space, providing them with generality and diversity. Meanwhile, this ensures that the algorithm can effectively search for the optimal value and avoid local minima. Therefore, Lévy flight was employed to update the position of the sand cat during the prey attack phase, and its mathematical model is as follows:

$$\begin{aligned} \overrightarrow{DD} &= \left| rand \cdot \overrightarrow{P_{best}(t)} - \overrightarrow{P(t)} \right| \\ \overrightarrow{P(t+1)} &= \overrightarrow{P_{best}(t)} - Levy \cdot \vec{\xi} \cdot \overrightarrow{DD} \cdot cos(\vartheta) \\ Levy &= random(size(dim)) \oplus L(step) \\ L(step) &\sim |step|^{-1-\gamma}, 0 < \gamma < 2 \end{aligned} \quad , \tag{25}$$

where γ is an exponent. $step$ is the step size, which is computed as follows:

$$\begin{aligned} step &= \propto / |b|^{1/\gamma}, \propto \sim N(0, \psi_{\propto}^2), b \sim N(0, \psi_b^2) \\ \psi_{\propto} &= \left[\frac{\Gamma(1+\gamma) \cdot sin(\pi \cdot \gamma/2)}{\Gamma((1+\gamma)/2) \cdot 2^{(\gamma-1)/2}} \right]^{1/\delta}, \psi_b = 1 \end{aligned} \quad , \tag{26}$$

where $\propto$ and b obey normal distribution.

3.1.2. Generation of simulation samples

The optimization performance of neural network-based approaches largely relies on the quality and quantity of samples. In terms of sample quantity, acquiring a sufficient number of real flight samples often comes at a high cost. Additionally, in terms of sample quality, it is difficult to guarantee that every step of the UAV's flight path is optimal. Therefore, this paper addresses these challenges by generating training samples through simulation, as illustrated in Fig. 3. Firstly, a three-dimensional flyable space FS is constructed, and multiple obstacles of varying sizes and shapes are randomly generated within the FS region according to Eq. (1). Different positions are randomly selected as the starting and target positions for UAV path planning. Secondly, IFDS is employed to plan the obstacle-avoidance path. In each step of the planning process, based on the relative state of the UAV to the nearest obstacle, the ESCSO is adopted to search for the optimal parameters of IFDS, such as δ, φ, and α. Subsequently, the parameters of IFDS are updated, and the UAV is flown to the planned waypoints. The specific process is formulated as follows:

(1) Environment information: In the flight region FS, based on the current position of the UAV, onboard sensors are used to detect nearby

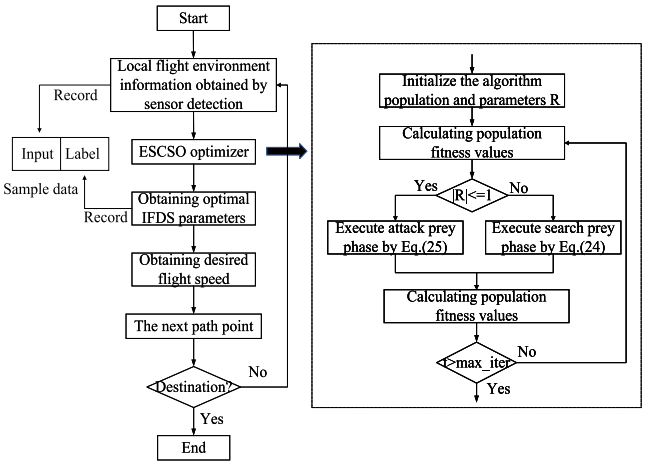


Fig. 3. The framework of simulation samples generation.

obstacle information and construct the local flight environment. At time step $\triangle T$, the current state information is recorded and its format is defined as follows:

$$input_T = (E - P_T, F - P_T, Obs_V^T) \quad . \tag{27}$$

(2) Initialization population: The parameters of IFDS, such as the repulsion coefficient, tangential coefficient, and tangential direction coefficient, are treated as individuals in the population and initialized based on their value ranges. The format of the population is defined as follows:

$$lable_i = \left[\delta_i, \varphi_i, \alpha_i \right], i \in [1, ZP] \quad . \tag{28}$$

(3) Fitness function: The fitness function value is employed to evaluate the quality of individuals in the population. During the sample generation process, individuals with low fitness are considered optimal individuals. Therefore, the fitness function designed in this paper is presented as follows:

$$Fit = fi + di + C_O + C_s \quad , \tag{29}$$

where fi is the planned path distance from the starting point to the current point, and di is the estimated distance from the current point to the destination. C_O and C_s refer to obstacle threat and path smoothness, respectively.

(4) Update operation: Based on Eqs. (24) and (25) of the ESCSO algorithm, the population individuals are updated. If the termination conditions of the algorithm are satisfied, the optimal value is output; Otherwise, the algorithm continues to be executed.

(5) Sample generation: At time step $\triangle T$, we use Eq. (27) as the input part of the sample and the optimal individual $lable_i^*$ obtained by the ESCSO algorithm optimization as the label of the sample. This creates a complete sample, which is recorded. Subsequently, the IFDS algorithm parameters are updated to obtain the next path point according to Eq. (13). The IFDS algorithm is judged to satisfy the termination condition. If it is satisfied, the algorithm concludes; otherwise, the sample generation process continues to be executed in the next time step.

3.2. ESCSO for training MLP

For traditional algorithms such as gradient-based methods, information about the gradient of the loss function with respect to the optimization variables is required to train the neural network. However, gradient-based methods can lead to sluggish network training and susceptibility to local minima, while heavily relying on initial parameters [41]. In contrast, population-based intelligent algorithms alleviate

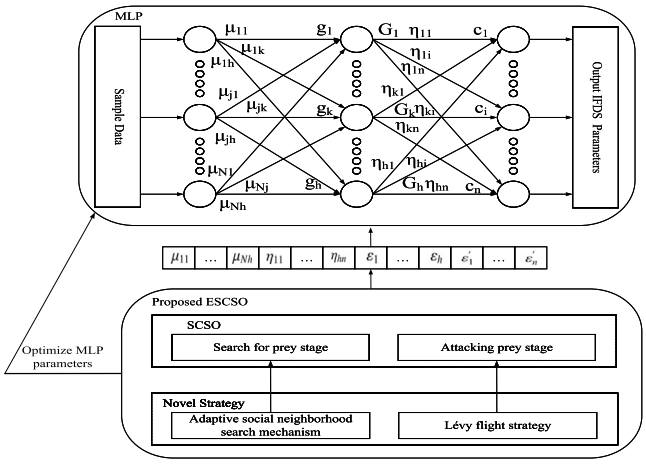


Fig. 4. The framework of ESCSO-based training MLP.

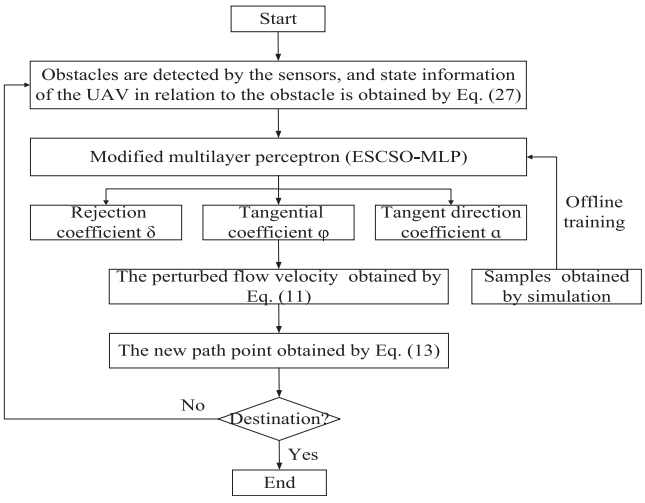


Fig. 5. The overall frame of real time path planning.

the need for gradient information and excel at circumventing local minima effectively. As depicted in Fig. 4, the purpose of this section is to utilize the ESCSO algorithm to train the MLP and obtain adaptive parameters for the IFDS algorithm in complex dynamic environments. To achieve this goal, two important issues need to be addressed: how to encode MLP parameters as scheme solutions within the ESCSO algorithm, and how to define the fitness function (also known as the loss function in MLP). In the ESCSO optimizer, each individual in the population is encoded as a one-dimensional vector, which consists of the weights connecting the input and hidden layers, weights linking the hidden and output layers, and the bias of each neuron. Each individual denotes a candidate solution for the MLP. Therefore, the dimension number of each individual in ESCSO is determined by $dim = Nh + hn + h + n$. The specific representation is shown below:

$$WB = \left[\underbrace{\mu_{11}, \ldots, \mu_{Nh}}_{input-hidden\ weights}, \underbrace{\eta_{11}, \ldots, \eta_{hn}}_{hidden-output\ weights}, \underbrace{\varepsilon_1, \ldots, \varepsilon_h}_{hidden\ biases}, \underbrace{\varepsilon'_1, \ldots, \varepsilon'_n}_{output\ biases} \right]. \quad (30)$$

Another crucial aspect is the setting of the fitness function, which is used to evaluate the quality of all individuals. In this paper, the mean-squared error (MSE) is employed as the fitness function of the ESCSO algorithm. In general, the smaller the value of MSE, the better the performance of the model. It is specifically defined as follows:

$$MSE = \frac{1}{num} \sum_{i=1}^{num} \sum_{j=1}^{n} \left(label_j - label'_j \right)^2, \quad (31)$$

where num indicates the number of training sample data; $label'$ is the actual output value of the MLP neural network; $label$ is the sample label value.

3.3. The framework for real time path planning

In this section, as shown in Fig. 5, we will present the operational steps for real-time path planning, and the specific process is as follows:

Step 1. Configuring the starting point and target point of the UAV. Based on the location of the UAV, the local flight environment is acquired through onboard sensors. Then, three state vectors are computed, including the vector from the UAV to the target point, the vector from the UAV to the nearest obstacle surface, and the velocity vector of the obstacle.

Step 2. The above three state vectors are fed into the trained MLP network as input data, which calculates the IFDS algorithm parameters.

Step 3. Based on the latest parameters, Eq. (13) is applied to calculate the UAV path position.

Table 1
The parameter settings of algorithms.

Algorithms	Parameter values
MAPF	step size $L = 0.1$
P-RRT	step size $S = 0.1$
TD3	$\gamma 1 = 0.99$, $t = 0.01$, $CR = 0.001$, $AR = 0.001$, $BS = 5000$
IFDS	$\delta = 1$, $\varphi = 1.5$, $\alpha = 1.5$
HHO-MLP-IFDS	$u \in (0,1)$, $v \in (0,2)$, $\beta 1 = 1.5$
SCSO-MLP-IFDS	$S = 2$
SOS-MLP-IFDS	$BF1 = 0$ or 1, $BF2 = 0$ or 1
MPA-MLP-IFDS	$FADs = 0.2$
WGWOIC-MLP-IFDS	$r1 \in [0,1]$, $r2 \in [0,1]$, $r3 \in [0,1]$
ESCSO-MLP-IFDS	$S = 2$

Step 4. Determine whether the planned path point has reached the destination. If not, steps 1–3 are continued. Otherwise, the algorithm is terminated.

4. Simulation experiment

To verify the usability of the proposed real-time path planning method in complex dynamic environments, a series of simulations and experiments have been conducted. In each scenario, the real-time path planning algorithm is applied to generate flight paths for the UAVs, aiming to validate the performance and effectiveness of the algorithm in different environments. By comparing the generated paths with those of other methods, we can evaluate the effectiveness, real-time capability, and security of the algorithm. Specific details are described as follows:

4.1. Simulation settings

To examine the effectiveness of the proposed algorithm, we constructed multiple obstacles of different sizes to form three complex flight scenarios by using different coefficients according to Eq. (1). Scenario 1 is a flight environment formed by several static obstacles constructed from cones and cylinders of different sizes. It is designed to validate the adaptive performance of the proposed algorithm when encountering different types of obstacles. Scenario 2 is a flight environment of dynamic obstacles formed by multiple spheres along different trajectories, which is mainly applied to test the adaptability of the proposed method to dynamic obstacles and the real-time and safety of path planning. Scenario 3 entails a complex flight environment that incorporates both the aforementioned dynamic and static obstacles, which mainly investigates the effect of the proposed algorithm

Table 2
Experimental results of all algorithms.

No.	Algorithms	Result				
		LS(deg)	GS(deg)	Num	Flight time(s)	Length(km)
1	MAPF	33.42	1.40	211	1266	21.22
	P-RRT	43.26	9.64	184	1104	18.37
	TD3	33.49	**1.28**	197	1182	19.58
	IFDS	88.74	3.57	307	1842	25.49
	HHO-MLP-IFDS	49.57	4.09	213	1278	21.46
	SCSO-MLP-IFDS	58.36	5.13	125	750	19.20
	SOS-MLP-IFDS	56.93	5.66	218	1308	20.32
	MPA-MLP-IFDS	48.32	5.89	114	684	17.88
	WGWOIC-MLP-IFDS	46.51	3.67	240	1440	21.81
	ESCSO-MLP-IFDS	**33.36**	5.99	**100**	**600**	**17.53**
2	MAPF	63.61	3.41	223	1338	22.35
	P-RRT	62.06	12.29	183	1098	18.51
	TD3	58.90	3.78	193	1158	21.81
	IFDS	60.07	5.12	315	1890	33.07
	HHO-MLP-IFDS	42.31	3.48	213	1278	20.43
	SCSO-MLP-IFDS	38.46	3.05	179	1074	19.53
	SOS-MLP-IFDS	41.94	4.66	216	1296	22.01
	MPA-MLP-IFDS	50.00	2.66	170	1020	18.50
	WGWOIC-MLP-IFDS	40.41	3.26	245	1470	21.18
	ESCSO-MLP-IFDS	**33.67**	**2.59**	**170**	**1020**	**18.40**
3	MAPF	27.11	1.82	200	1200	20.21
	P-RRT	45.73	8.28	174	1044	17.30
	TD3	26.70	**1.36**	189	1134	18.76
	IFDS	68.06	2.56	297	1782	24.46
	HHO-MLP-IFDS	42.45	3.87	124	744	17.77
	SCSO-MLP-IFDS	47.47	5.16	113	678	18.29
	SOS-MLP-IFDS	51.97	3.89	140	840	19.00
	MPA-MLP-IFDS	47.00	4.94	100	600	17.37
	WGWOIC-MLP-IFDS	43.30	4.02	142	852	18.06
	ESCSO-MLP-IFDS	**26.61**	4.91	**99**	**594**	**17.19**

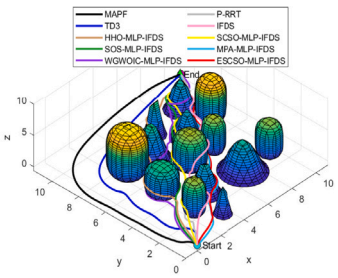
when dealing with both dynamic and static obstacles in a complex environment. In the first flight environment, the starting position of the UAV is $(0, 0, 0.3)$ and the ending position is $(11, 11, 0.4)$. In the second flight environment, the starting position of the UAV is $(0, 0, 4)$ and the ending position is $(15, 7, 6)$. In the third flight environment, the starting position of the UAV is $(0, 0, 0.9)$ and the ending position is $(11, 11, 1.4)$. In the evaluation, the ESCSO for training the MLP is also compared with other advanced optimizers such as HHO [42], SCSO [32], WGWOIC [43], SOS [44], and MPA [45]. The population size for all algorithms is 30 and the total number of iterations is set to 700. In addition, the proposed algorithm is also compared with existing methods such as MAPF [46], P-RRT [47], and TD3 [48] to validate the effectiveness of the algorithm. The critical parameters of all the algorithms are listed in Table 1. All experiments and simulations were conducted using MATLAB R2021(a) on a computer equipped with an Intel Core i5-10400 processor and 16 GB of memory. In this work, we follow Kolmogorov's theorem, which shows that a three-layer MLP can approximate any continuous function [49,50]. This implies that in the experiments of this study, one hidden layer is sufficient. Therefore, the MLP structure used in the proposed approach included three layers: one input layer, one hidden layer, and one output layer. The number of neurons for the input layer, hidden layer, and output layer are set to 9, 150, and 3, respectively. The information transfer in the MLP is primarily achieved through Eqs. (14)–(16). Meanwhile, the sampling period $\triangle T$ of the IFDS algorithm is set to 0.1.

4.2. Experiment findings

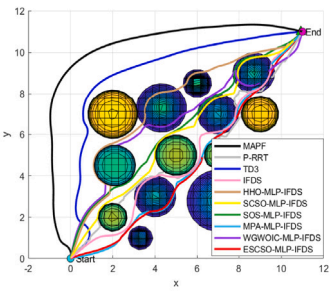
Table 2 presents the numerical simulation results for three complex scenarios, as well as the optimal values of path maximum tortuosity (LS), average tortuosity (GS), number of waypoints (Num), flight time, and path length obtained when each algorithm is executed independently 20 times. The optimal results are highlighted in bold. Figs. 6, 7, and 8 illustrate the flight status of the UAV in each of the three scenarios.

4.2.1. Scenario 1

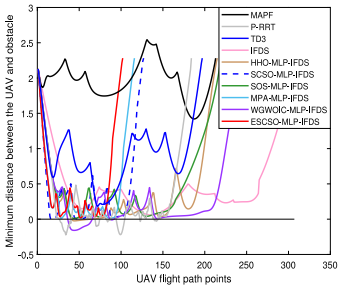
For the first scenario, we constructed a dense static obstacle environment with various shapes and sizes to validate the real-time and effectiveness of the proposed algorithm for UAV path planning. Scenario 1 in Table 2 presents the experimental outcomes for the first static test setting. From the numerical results, it can be observed that the obstacle avoidance paths planned for the UAV by the ESCSO-MLP-IFDS algorithm get better maximum tortuosity, number of path points, flight time, and path length than the values of the other comparative methods. These comparative metrics demonstrate that the proposed method has smaller LS smoothness values, fewer path points, shorter flight time, and smaller path lengths. Compared to the IFDS method with fixed parameter values, the proposed method with adaptively varying parameter values is more capable of solving real-time path planning problems in complex environments. Fig. 6 presents more visually the superiority of the proposed algorithm. Fig. 6a illustrates the different findings of the ten algorithms for path planning in 3D environments. The performance of the algorithms can be effectively verified because the environment model has many dense and different-sized obstacles. According to the simulation results, compared with the proposed algorithms, the flight paths obtained by the algorithms are more tortuous, except for the MAPF, TD3, and MPA-MLP-IFDS algorithms. The top view of the paths in Fig. 6b also indicates that the paths generated by other algorithms not only have longer lengths but also result in increased flight time and fuel consumption. In contrast, the proposed algorithm demonstrates advantages in achieving the smoothest and shortest paths. The distances between the paths planned by all algorithms and obstacles are shown in Fig. 6c. From the graph, it can be observed that except for the P-RRT, WGWOIC-MLP-IFDS, and HHO-MLP-IFDS algorithms, which cannot provide effective paths, all other algorithms are capable of providing safe flight paths for the UAV. Fig. 6(d∼f) exhibits the real-time paths planned by the proposed algorithm at different time steps based on the local obstacle information obtained from the onboard sensor detection. From the graph, it can be observed that the proposed algorithm can provide feasible flight paths



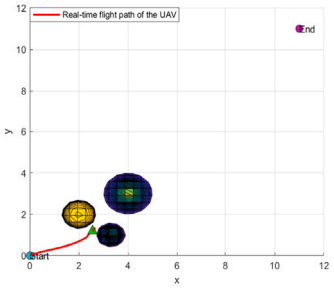
(a) 3D path planning for all algorithms



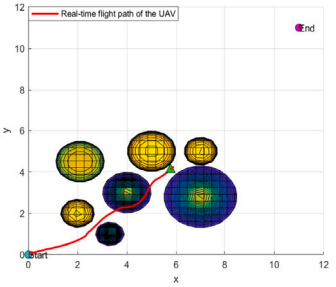
(b) Top view of the paths for all algorithms



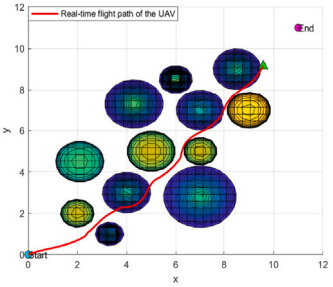
(c) Distance between UAV and the surface of the nearest obstacle



(d) Real-time planning results of the proposed algorithm at flight time 96s

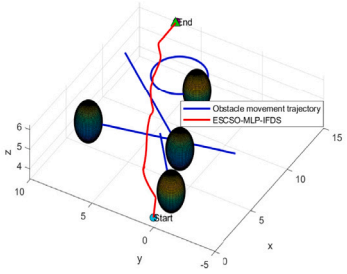


(e) Real-time planning results of the proposed algorithm at flight time 264s

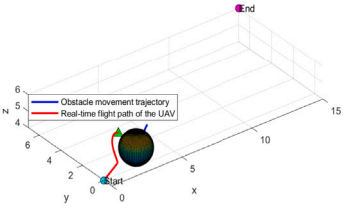


(f) Real-time planning results of the proposed algorithm at flight time 498s

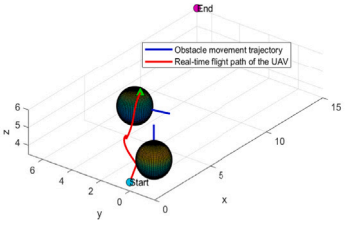
Fig. 6. Flight situation of UAV in Scenario 1.



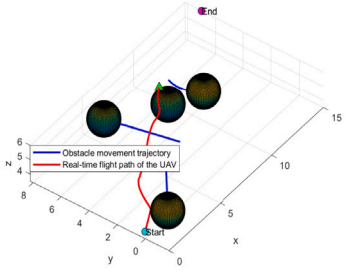
(a) 3D path planning obtained by the proposed algorithm



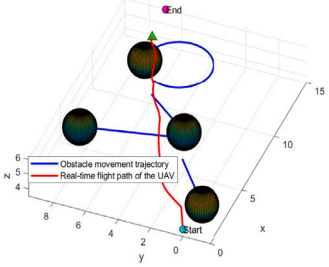
(b) Real-time planning results of the proposed algorithm at flight time 240s



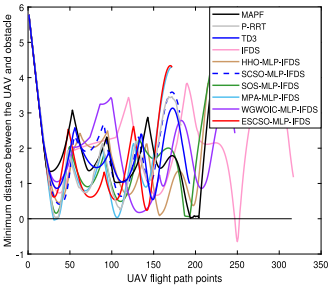
(c) Real-time planning results of the proposed algorithm at flight time 444s



(d) Real-time planning results of the proposed algorithm at flight time 612s



(e) Real-time planning results of the proposed algorithm at flight time 864s



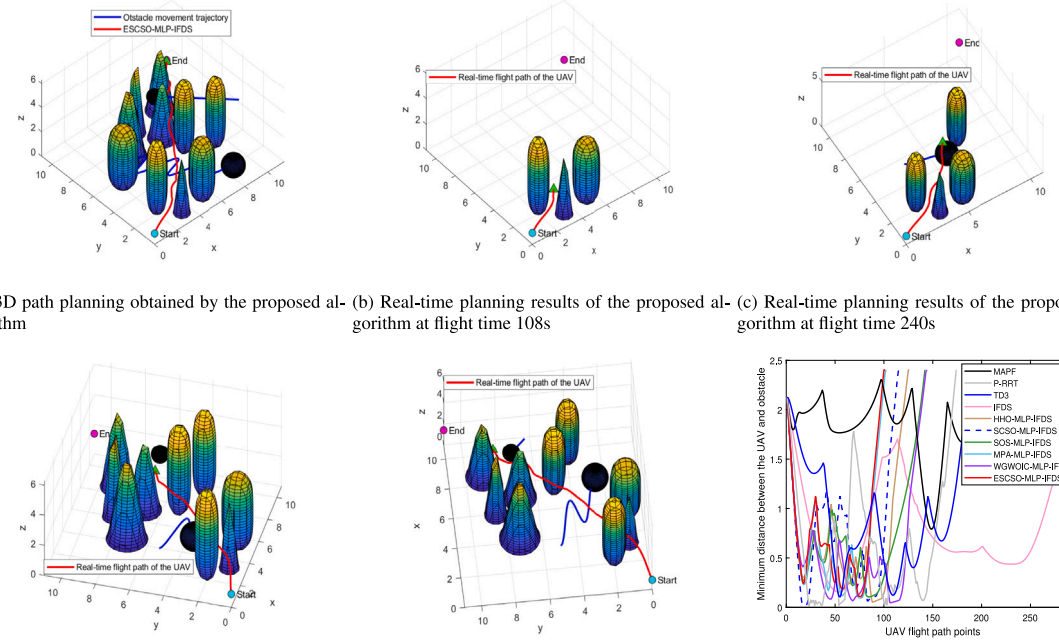
(f) Distance between UAV and the surface of the nearest obstacle

Fig. 7. Flight situation of UAV in Scenario 2.

for the UAV in complex unknown environments, satisfying constraints such as path length, sudden threats, turning angles, and climbing angles, based on real-time dynamic information.

4.2.2. Scenario 2

In this section, we designed a dynamic test scenario for evaluating the performance of the proposed algorithm in a dynamic environment.



(a) 3D path planning obtained by the proposed algorithm

(b) Real-time planning results of the proposed algorithm at flight time 108s

(c) Real-time planning results of the proposed algorithm at flight time 240s

(d) Real-time planning results of the proposed algorithm at flight time 336s

(e) Real-time planning results of the proposed algorithm at flight time 438s

(f) Distance between UAV and the surface of the nearest obstacle

Fig. 8. Flight situation of UAV in Scenario 3.

The scenario includes four spheres acting as dynamic obstacles, and they move with time, coming from different directions to simulate a complex dynamic environment. The proposed algorithm is executed for this dynamic scenario, and the experimental results are recorded. Subsequently, the simulation path planning results of the algorithm at each time step are presented, including the trajectory of the UAV and the motion trajectories of dynamic obstacles. The details are shown in Fig. 7. Fig. 7a presents the final planned path provided by the proposed method for the UAV from the starting point to the destination. For a clear view of the process, Fig. 7b–Fig. 7e display more detailed real-time UAV obstacle avoidance processes when obstacles approach the UAV. From Fig. 7b, it is evident that in the initial stage, the UAV is flying towards the destination. When the onboard sensor detects dynamic obstacles approaching from the opposite direction, the navigation direction is adjusted in advance, and the UAV easily maneuvers to the left to bypass the obstacles. In Fig. 7c, the UAV encounters a dynamic obstacle moving from right to left. Based on the sensor detection information, the proposed algorithm effectively avoids collision by intelligently adjusting the climbing angle, thereby ensuring a safe distance between the UAV and the obstacle. Similarly, in Fig. 7d and Fig. 7e, we can observe that the proposed algorithm has the capability to dynamically adjust the UAV's flight path based on real-time environmental information, enabling navigation in complex environments and avoiding potential collisions with moving obstacles. This ability is crucial for the safe and reliable operation of UAVs. The distance between the UAV and the dynamic threat surface is displayed in Fig. 7f, from which it can be gotten that except for the MPA-MLP-IFDS and IFDS algorithms, the minimum distances between the UAV and the dynamic threat surfaces are all greater than 0. This indicates that the UAV is able to successfully avoid dynamic threats and reach its destination. In addition, we also conducted a quantitative analysis of the algorithm's performance in the dynamic testing scenario. We examined metrics such as maximum tortuosity and average tortuosity of path planning, number of path points, flight time, and path length, and compared them with other comparison algorithms. From Scenario 2 in Table 2, it can be observed that the proposed algorithm ranks first in all the mentioned metrics. Although the MPA-MLP-IFDS algorithm obtains the same number of path points and flight times as the proposed algorithm, the paths generated by the MPA-MLP-IFDS algorithm are ineffective and unsafe. In summary, our proposed dynamic obstacle avoidance algorithm has great potential in tackling challenges in dynamic scenarios. It provides a reliable path planning solution for UAVs to execute missions in complex and dynamic environments.

4.2.3. Scenario 3

In the third scenario, we designed a mixed dynamic-static environment by combining elements from Scenario 1 and Scenario 2 to simulate a complex flight environment, aiming to further evaluate the effectiveness of the proposed method for UAV path planning. This scenario includes two moving obstacles: Obstacle 1 with a threat radius of 0.8 executing wave-like movements, and Obstacle 2 with a threat radius of 0.5 moving along a linear trajectory. The experimental results of this flight environment are recorded in Scenario 3 of Table 2, and through quantitative analysis, we can obtain that the proposed algorithm outperformed the comparison algorithms in terms of maximum tortuosity, number of path points, flight time, and path length. Although the values obtained by the proposed algorithm are not as good as the values obtained by other algorithms in terms of average tortuosity, the values obtained by the ESCSO-MLP-IFDS algorithm are very competitive. These results demonstrate that the proposed approach is capable of generating efficient paths that satisfy constraints and minimize flight time. Fig. 8 showcases the real-time path planning results for the UAV in this dynamic-static environment. Fig. 8a displays the complete flight path planned by the proposed algorithm for the UAV. From the graph, it can be observed that the UAV can safely follow the planned path and reach its destination without any collisions. Fig. 8b vividly demonstrates the effectiveness of the proposed algorithm, which dynamically and efficiently adjusts the UAV's path in response to detected static obstacles, ensuring that the UAV can avoid potential collisions. Fig. 8c graphically demonstrates the dynamic responsiveness of the proposed algorithm to the changing environment, enabling the UAV to successfully navigate the UAV by intelligently adjusting the pitch angle to avoid dynamic obstacles. Fig. 8d and Fig. 8e depicts the presence of various static obstacles of different sizes and dynamic obstacle within the UAV's sensor detection range. The proposed real-time planning algorithm not only provides a safe and smooth path for the UAV to

navigate through narrow passages but also enables the UAV to perform effective maneuvering actions, such as making turns, to avoid dynamic threats. Meanwhile, Fig. 8f also exhibits that the paths planned by all the algorithms are able to maintain a safe flight distance for the UAV from the obstacles. These results highlight the versatility and robustness of the proposed algorithm, making it a reliable solution for real-time path planning in complex and dynamic environments for UAV missions.

5. Discussion analysis

In this section, through a comprehensive analysis of the experimental results and algorithm characteristics, we conducted an evaluation and explanation of the proposed real-time path planning algorithm. The superiority of the algorithm mainly comes from the following aspects:

Firstly, the algorithm ingeniously combines the advantages of IFDS and ANN, which can make full use of the dynamic environment information and sensor detection data, and intelligently adjusts the IFDS parameters through the ANN, so as to provide real-time and accurate path planning guidance for UAVs. Compared with other methods, the proposed algorithm shows stronger adaptability and robustness under high real-time requirements.

Secondly, this study designs complex dynamic and static simulation scenarios to generate a large number of high-quality training samples for supporting neural network training. The application of the enhanced sand cat swarm optimization algorithm (ESCSO) further enhances the sample quality, addressing the issue of neural networks relying on high-quality samples. The above measures have resulted in significant adaptability of the proposed algorithm to effectively deal with various environments and obstacle types. It enables the algorithm to perform well in the face of unpredictable changes, such as moving obstacles. In contrast, other path planning algorithms may perform poorly in dynamic situations.

Thirdly, the ESCSO algorithm is applied to optimize the parameters of the MLP, thereby enhancing the neural network's ability to accurately predict environmental state information. This provides a solid foundation for the algorithm's real-time and precise performance. Furthermore, the proposed IFDS-based algorithm can simultaneously consider all detected obstacles through disturbance matrices without the need for additional checks to determine obstacle avoidance. This streamlined approach allows the algorithm to adapt to evolving environments with minimal computational resources, making it more concise and efficient.

Finally, the IFDS-based algorithm employs a fluid dynamics model to generate smooth paths that conform to the constraints of UAV flight, enabling efficient obstacle avoidance. In contrast, other algorithms often produce paths with abrupt changes that do not satisfy the necessary constraints, requiring additional computational resources to address path smoothness. This may lead to a lack of real-time performance and computational inefficiency.

6. Conclusions and future directions

In this study, a real-time path planning algorithm is proposed to address the problem of efficient, safe, and reliable path planning for UAVs in complex environments. Firstly, the algorithm effectively combines the powerful adaptive learning capability of MLP with the efficient obstacle avoidance capability of IFDS, enabling intelligent adaptation to dynamic and static environmental changes and providing excellent path guidance for UAVs. Secondly, by employing ESCSO to generate simulation samples, the issue of low-quality training samples for MLP is effectively addressed. Meanwhile, ESCSO is also employed to optimize network parameters, which effectively solves the difficulty that the traditional methods have fallen into local optimum. Finally, experimental results from three scenarios demonstrate that the ESCSO-MLP-IFDS algorithm enables UAVs to make accurate and safe flight decisions in real-time demanding situations.

However, despite the achievements of the proposed algorithm, we also recognize that there is room for further improvement. In the future, we need to further optimize and adjust the algorithm to enhance its ability to handle large-scale dynamic obstacles when facing more complex and dense dynamic environments.

CRediT authorship contribution statement

Yanbiao Niu: Conceptualization, Methodology, Software, Validation, Writing – original draft. **Xuefeng Yan:** Formal analysis, Funding acquisition, Project administration, Resources, Supervision, Writing – review & editing. **Yongzhen Wang:** Investigation, Writing – review & editing. **Yanzhao Niu:** Investigation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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References

[1] S. Aslan, T. Erkin, A multi-population immune plasma algorithm for path planning of unmanned combat aerial vehicle, Adv. Eng. Inform. 55 (2023) 101829.
[2] W. Yi, M. Sutrisna, H. Wang, Unmanned aerial vehicle based low carbon monitoring planning, Adv. Eng. Inform. 48 (2021) 101277.
[3] S.-H. Huang, Y.-H. Huang, C.A. Blazquez, C.-Y. Chen, Solving the vehicle routing problem with drone for delivery services using an ant colony optimization algorithm, Adv. Eng. Inform. 51 (2022) 101536.
[4] F. Santoso, M.A. Garratt, S.G. Anavatti, State-of-the-art intelligent flight control systems in unmanned aerial vehicles, IEEE Trans. Autom. Sci. Eng. 15 (2) (2017) 613–627.
[5] Y. Niu, X. Yan, Y. Wang, Y. Niu, Three-dimensional UCAV path planning using a novel modified artificial ecosystem optimizer, Expert Syst. Appl. 217 (2023) 119499.
[6] X. Pu, C. Zhang, J. Zhang, Multi-target real-time path planning using double adaptive a algorithm, IEEE Trans. Aerosp. Electron. Syst. (2023) 1–11.
[7] S.-K. Huang, W.-J. Wang, C.-H. Sun, A new multirobot path planning with priority order based on the generalized Voronoi diagram, IEEE Access 10 (2022) 56564–56577.
[8] M. Kothari, I. Postlethwaite, A probabilistically robust path planning algorithm for UAVs using rapidly-exploring random trees, J. Intell. Robot. Syst. 71 (2) (2013-08) 231–253.
[9] S. Moon, E. Oh, D.H. Shim, An integral framework of task assignment and path planning for multiple unmanned aerial vehicles in dynamic environments, J. Intell. Robot. Syst. 70 (1–4) (2013) 303–313.
[10] Y. bo Chen, G. chen Luo, Y. song Mei, J. qiao Yu, X. long Su, UAV path planning using artificial potential field method updated by optimal control theory, Internat. J. Systems Sci. 47 (6) (2016) 1407–1420.
[11] Q. Su, S. Ma, L. Wang, Y. Song, H. Wang, B. Li, Y. Yang, Artificial potential field guided JPS algorithm for fast optimal path planning in cluttered environments, J. Braz. Soc. Mech. Sci. Eng. 44 (12) (2022) 602.
[12] Z. Wu, S. Dong, M. Yuan, J. Cui, L. Zhao, C. Tong, Rotate artificial potential field algorithm toward 3D real-time path planning for unmanned aerial vehicle, Proc. Inst. Mech. Eng. G 237 (4) (2023) 940–955.
[13] U. Goel, S. Varshney, A. Jain, S. Maheshwari, A. Shukla, Three dimensional path planning for UAVs in dynamic environment using glow-worm swarm optimization, Procedia Comput. Sci. 133 (2018) 230–239, International Conference on Robotics and Smart Manufacturing (RoSMa2018).
[14] B. Zhang, H. Duan, Three-dimensional path planning for uninhabited combat aerial vehicle based on predator-prey pigeon-inspired optimization in dynamic environment, IEEE/ACM Trans. Comput. Biol. Bioinform. 14 (1) (2017) 97–107.
[15] Real-time path planning for autonomous vehicle based on teaching–learning-based optimization, 15 (3) (2022) 381–398.

[16] S. Hosseininejad, C. Dadkhah, Mobile robot path planning in dynamic environment based on cuckoo optimization algorithm, Int. J. Adv. Robot. Syst. 16 (2) (2019) 172988141983957.
[17] J. Chen, P. Han, Y. Zhang, T. You, P. Zheng, Scheduling energy consumption-constrained workflows in heterogeneous multi-processor embedded systems, J. Syst. Archit. 142 (2023) 102938.
[18] J. Chen, Y. Zhang, L. Wu, T. You, X. Ning, An adaptive clustering-based algorithm for automatic path planning of heterogeneous UAVs, IEEE Trans. Intell. Transp. Syst. 23 (9) (2022) 16842–16853.
[19] J. Chen, F. Ling, Y. Zhang, T. You, Y. Liu, X. Du, Coverage path planning of heterogeneous unmanned aerial vehicles based on ant colony system, Swarm Evol. Comput. 69 (2022) 101005.
[20] C. Ye, N. Yung, D. Wang, A fuzzy controller with supervised learning assisted reinforcement learning algorithm for obstacle avoidance, IEEE Trans. Syst. Man Cybern. B 33 (1) (2003) 17–27.
[21] N. Zhang, M. Zhang, K.H. Low, 3D path planning and real-time collision resolution of multirotor drone operations in complex urban low-altitude airspace, Transp. Res. C 129 (2021) 103123.
[22] P. Chen, J. Pei, W. Lu, M. Li, A deep reinforcement learning based method for real-time path planning and dynamic obstacle avoidance, Neurocomputing 497 (2022) 64–75.
[23] J. Chen, T. Li, Y. Zhang, T. You, Y. Lu, P. Tiwari, N. Kumar, Global-and-local attention-based reinforcement learning for cooperative behaviour control of multiple UAVs, IEEE Trans. Veh. Technol. (2023) 1–13.
[24] C. Lin, H. Wang, J. Yuan, D. Yu, C. Li, Research on UUV obstacle avoiding method based on recurrent neural networks, Complexity 2019 (2019) 1–16.
[25] H. Zhu, F.M. Claramunt, B. Brito, J. Alonso-Mora, Learning interaction-aware trajectory predictions for decentralized multi-robot motion planning in dynamic environments, IEEE Robot. Autom. Lett. 6 (2) (2021) 2256–2263.
[26] Y. Liu, Z. Zheng, F. Qin, X. Zhang, H. Yao, A residual convolutional neural network based approach for real-time path planning, Knowl.-Based Syst. 242 (2022) 108400.
[27] J. Wu, H. Wang, N. Li, P. Yao, Y. Huang, H. Yang, Path planning for solar-powered UAV in urban environment, Neurocomputing 275 (2018) 2055–2065.
[28] H. Wang, W. Lyu, P. Yao, X. Liang, C. Liu, Three-dimensional path planning for unmanned aerial vehicle based on interfered fluid dynamical system, Chin. J. Aeronaut. 28 (1) (2015) 229–239.
[29] R. Zhao, Y. Wang, P. Hu, H. Jelodar, C. Yuan, Y. Li, I. Masood, M. Rabbani, Selfish herds optimization algorithm with orthogonal design and information update for training multi-layer perceptron neural network, Appl. Intell. 49 (2019) 2339–2381.
[30] C.H. Goay, A. Abd Aziz, N.S. Ahmad, P. Goh, Eye diagram contour modeling using multilayer perceptron neural networks with adaptive sampling and feature selection, IEEE Trans. Compon. Packag. Manuf. Technol. 9 (12) (2019) 2427–2441.
[31] K. Sun, S.-H. Huang, D.S.-H. Wong, S.-S. Jang, Design and application of a variable selection method for multilayer perceptron neural network with LASSO, IEEE Trans. Neural Netw. Learn. Syst. 28 (6) (2016) 1386–1396.
[32] A. Seyyedabbasi, F. Kiani, Sand cat swarm optimization: A nature-inspired algorithm to solve global optimization problems, Eng. Comput. (2022) 1–25.
[33] J. Wu, H. Wang, N. Li, Z. Su, Formation obstacle avoidance: A fluid-based solution, IEEE Syst. J. 14 (1) (2020) 1479–1490.
[34] R. Poli, J. Kennedy, T. Blackwell, Particle swarm optimization, Swarm Intell. 1 (1) (2007) 33–57.
[35] S. Mirjalili, S.M. Mirjalili, A. Lewis, Grey wolf optimizer, Adv. Eng. Softw. 69 (2014) 46–61.
[36] S. Mirjalili, A.H. Gandomi, S.Z. Mirjalili, S. Saremi, H. Faris, S.M. Mirjalili, Salp swarm algorithm: A bio-inspired optimizer for engineering design problems, Adv. Eng. Softw. 114 (2017) 163–191.
[37] R. Zhao, Y. Wang, C. Liu, P. Hu, Y. Li, H. Li, C. Yuan, Selfish herd optimizer with levy-flight distribution strategy for global optimization problem, Physica A 538 (2020) 122687.
[38] Y. Ling, Y. Zhou, Q. Luo, Lévy flight trajectory-based whale optimization algorithm for global optimization, IEEE Access 5 (2017) 6168–6186.
[39] C.T. Brown, L.S. Liebovitch, R. Glendon, Lévy flights in Dobe Ju/'hoansi foraging patterns, Hum. Ecol. 35 (2007) 129–138.
[40] I. Pavlyukevich, Lévy flights, non-local search and simulated annealing, J. Comput. Phys. 226 (2007) 1830–1844.
[41] Q.-V. Pham, T. Huynh-The, M. Alazab, J. Zhao, W.-J. Hwang, Sum-rate maximization for UAV-assisted visible light communications using NOMA: Swarm intelligence meets machine learning, IEEE Internet Things J. 7 (10) (2020) 10375–10387.
[42] S. Ohadi, S.A. Hashemi Monfared, M. Azhdary Moghaddam, M. Givehchi, Feasibility of a novel predictive model based on multilayer perceptron optimized with Harris hawk optimization for estimating of the longitudinal dispersion coefficient in rivers, Neural Comput. Appl. 35 (9) (2023) 7081–7105.
[43] A. Kumar, A. Kumar, et al., Weighted grey wolf optimizer with improved convergence rate in training multi-layer perceptron to solve classification problems, Jordanian J. Comput. Inf. Technol. 7 (3) (2021).
[44] S. Rastbod, F. Rahimi, Y. Dehghan, S. Kamranfar, O. Benjeddou, M.L. Nehdi, An optimized machine learning approach for forecasting thermal energy demand of buildings, Sustainability 15 (1) (2022) 231.
[45] Z. Dong, Z. Sheng, Y. Zhao, P. Zhi, Robust optimization design method for structural reliability based on active-learning MPA-BP neural network, Int. J. Struct. Integr. 14 (2) (2023) 248–266.
[46] S.M.H. Rostami, A.K. Sangaiah, J. Wang, X. Liu, Obstacle avoidance of mobile robots using modified artificial potential field algorithm, EURASIP J. Wirel. Commun. Netw. 2019 (2019) 70.
[47] Z. Zhang, B. Qiao, W. Zhao, X. Chen, A predictive path planning algorithm for mobile robot in dynamic environments based on rapidly exploring random tree, Arab. J. Sci. Eng. 46 (2021) 8223–8232.
[48] S. Zhang, Y. Li, Q. Dong, Autonomous navigation of UAV in multi-obstacle environments based on a deep reinforcement learning approach, Appl. Soft Comput. 115 (2022) 108194.
[49] Y. Wang, L. Wang, Q. Chang, C. Yang, Effects of direct input–output connections on multilayer perceptron neural networks for time series prediction, Soft Comput. 24 (2020) 4729–4738.
[50] K. Hornik, M. Stinchcombe, H. White, Multilayer feedforward networks are universal approximators, Neural Netw. 2 (5) (1989) 359–366.